

Andrea Maioli

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Experience

Freepoint Commodities

DATA SCIENTIST

Zug, Switzerland

Sep. 2024 - Present

- **Country-level power forecasting:** forecasted country level wind and solar energy production, focusing on coverage calibration and probabilistic methods. Improved upon external vendor forecasts for the 7–15 day horizon, reducing MAE by 8%.
- **Data infrastructure:** led a cross-functional team (meteorologists + IT) to design and deliver a scalable weather datalake (zarr/parquet-based), streamlining data accessibility for analytics.
- **Software standardization:** developed and maintained core Python libraries adopted by cross-functional teams, standardizing data access and reducing onboarding times for new analysts.
- **Strategic cost savings:** built a performance dashboard enabling traders to benchmark vendor accuracy across various lead times and metrics. This led to the cancellation of sub-optimal subscriptions, saving the company ~€100k/y.

DXT Commodities SA

DATA SCIENTIST

Lugano, Switzerland

Mar. 2019 - Jul. 2024

- **Multi-asset forecasting:** forecasted 4Gw renewable portfolio power generation (wind, solar, hydro, consumption), power and gas demand of EU countries.
- **Anomaly detection:** developed an automatic outage and outlier detection for dispatched power plants.
- **Algorithmic trading:** designed and implemented an intraday trading strategy for the Italian power market (up to 6 hours before delivery), achieving a Sharpe ratio of 2.1.
- **MLOps architecture:** Built the end-to-end MLOps infrastructure from the ground up, integrating Cookiecutter, MLflow, Jenkins, and Prefect to automate the model lifecycle.

UniCredit S.p.a

QUANTITATIVE RISK ANALYST IN GROUP FINANCIAL RISK METHODOLOGIES

Milan, Italy

Apr. 2017 - Feb. 2019

Education

Università di Bologna

MASTER IN MATHEMATICAL FINANCE "CORSO ALTA FORMAZIONE FINANZA MATEMATICA"

Specialisation in Stochastic equations and Option pricing. Application of Montecarlo, quasi-Montecarlo and Fourier transform methods.

Bologna, Italy

Paris Saclay

MASTÉR M2 ANALYSE, MODÉLISATION ET SIMULATION - DISPERSIVE EQUATIONS

Thesis: Structure of global solutions for a damped Klein-Gordon equation.

Paris, France

Università di Pisa

BACHELOR DEGREE FIRST YEAR IN MATHEMATICS

Thesis: Cauchy-Kovaleskaya's theorem

Pisa, Italy

Tech Skills

Programming languages	Python (sklearn, torch, pygam, pandas polars, cvxpy, ONNX), SQL
Dashboard	Marimo (♥), Streamlit
MLOps	Git, MLflow, Docker, Jenkins, cookiecutter, Prefect, FastAPI
Models	GAM (♥), Catboost, Ensemble NeuralNet, NHITS, Conformal predictions
Cloud computing	AWS

Languages

Mother tongue	Italian
Fluent	English
Beginner	Swedish, French

Personal projects

pimplot	make matplotlib great again
lpspline	nice api for linear spline based model
bessopt	me learning OR by optimising BESS

Hobbies

Drummer	Funky, Indie rock
Board Games	Power Grid, Bang! (best player in the world)
Opera	La Cenerentola, Gianni Schicchi
Sports	Boxe, Basketball, Warhammer 40k

Dreams

Dunk at basketball	Work in progress... almost there...
Skateboard	Ollie to get on sidewalk.
Bake bread	Failed attempts: 5 — :-(✓
Kungsliden	Complete full Kungsliden trail. ✓